

TidyTuesday Week 40

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Monday 28th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

This week our tireless random-number generator reached deep into the FRED grab bag and produced a pairing so inevitable, so obviously destined, that historians will surely study it for generations: the real GDP of wood product manufacturing in the Great Lakes region versus the number of mobile cellular subscriptions per 100 people across Europe and Central Asia. Lumber in Michigan, phones in Kazakhstan. If you don't feel these two quantities were always meant to be together, I frankly don't know what to tell you.

Here's where it gets deliciously backwards. The raw regression – the one where we just fling the two series at each other and see if they stick – came back with a shrug. A p-value of 0.14, an R-squared of eight percent, and a footnote politely warning us about "numerical problems," which is the statistical equivalent of the software backing slowly out of the room. Normally this is the part where a shared trend swoops in to manufacture a fake romance, but no such luck: even the trend mirage couldn't be bothered to show up.

And then, plot twist, the detrended version – the one that strips out each series' own long-run drift and is supposed to be the harder, more honest test – actually got MORE significant. P-value drops to 0.02, R-squared doubles to about eighteen percent, and the coefficient stays stubbornly negative: more phones, less Great Lakes plywood. This is the rare inversion of our usual scandal. Instead of a raw hit collapsing under scrutiny, we have a nothingburger that firmed up once we removed the trends. Which, in the strange arithmetic of this project, makes it one of the more "interesting" results we've stumbled into, if you squint and lower your standards to floor level.

Now, before anyone drafts a theory about smartphones single-handedly bulldozing Midwestern sawmills, let me remind you with great affection: these two series were paired by a coin flip, there is no plausible mechanism connecting them, and eighteen percent of variance "explained" is still eighty-two percent of variance doing whatever it pleases. Spurious correlation is not a bug of this project, it is the entire point. Enjoy the number, cherish it, and please do not put it in a spreadsheet you send to your boss.

2 Date: 2026-09-28

Series ID: GLAKWPMANRGSP

This series is titled Real Gross Domestic Product: Wood Product Manufacturing (321) in the Great Lakes BEA Region and has a frequency of Annual. The units are Millions of Chained 2017 Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: ITCELSETSP2ECS

This series is titled Mobile Cellular Subscriptions: All Income Levels in Europe and Central Asia and has a frequency of Annual. The units are Number per 100 People and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1960-01-01 and the observation end date is 2025-01-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ITCELSETSP2ECS	R-squared:	0.083
Model:	OLS	Adj. R-squared:	0.048
Method:	Least Squares	F-statistic:	2.357
Date:	Mon, 28 Sep 2026	Prob (F-statistic):	0.137
Time:	18:50:21	Log-Likelihood:	-142.11
No. Observations:	28	AIC:	288.2
Df Residuals:	26	BIC:	290.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	285.3207	123.888	2.303	0.030	30.665	539.976
value_fred_GLAKWPMANRGSP	-0.0407	0.026	-1.535	0.137	-0.095	0.014

Omnibus:	3.492	Durbin-Watson:	0.174
Prob(Omnibus):	0.174	Jarque-Bera (JB):	2.983
Skew:	-0.713	Prob(JB):	0.225
Kurtosis:	2.275	Cond. No.	7.63e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 7.63e+04. This might indicate that there are strong multicollinearity or other numerical problems.

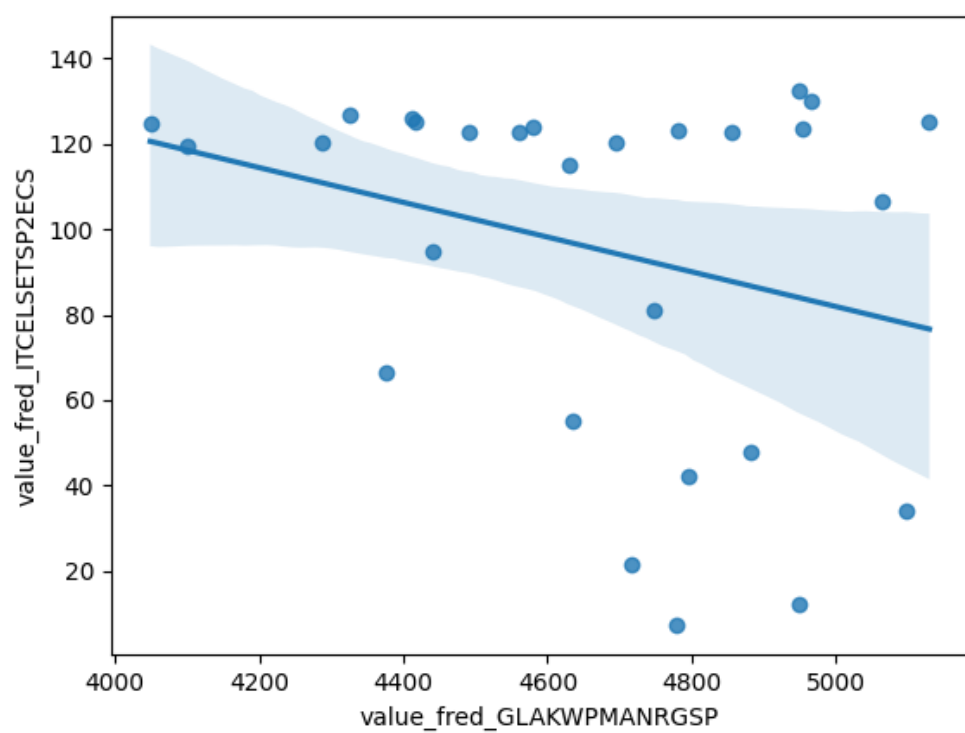


Figure 1: Raw Regression Plot for 2026-09-28

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ITCELSETSP2ECS_detrended	R-squared:	0.184
Model:	OLS	Adj. R-squared:	0.152
Method:	Least Squares	F-statistic:	5.853
Date:	Mon, 28 Sep 2026	Prob (F-statistic):	0.0228
Time:	18:50:21	Log-Likelihood:	-119.03
No. Observations:	28	AIC:	242.1
Df Residuals:	26	BIC:	244.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.116e-12	3.330	3.35e-13	1.000	-6.845	6.845
value_fred_GLAKWPMANRGSP_detrended	-0.0283	0.012	-2.419	0.023	-0.052	-0.004

Omnibus:	0.861	Durbin-Watson:	0.444
Prob(Omnibus):	0.650	Jarque-Bera (JB):	0.837
Skew:	0.216	Prob(JB):	0.658
Kurtosis:	2.272	Cond. No.	285.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

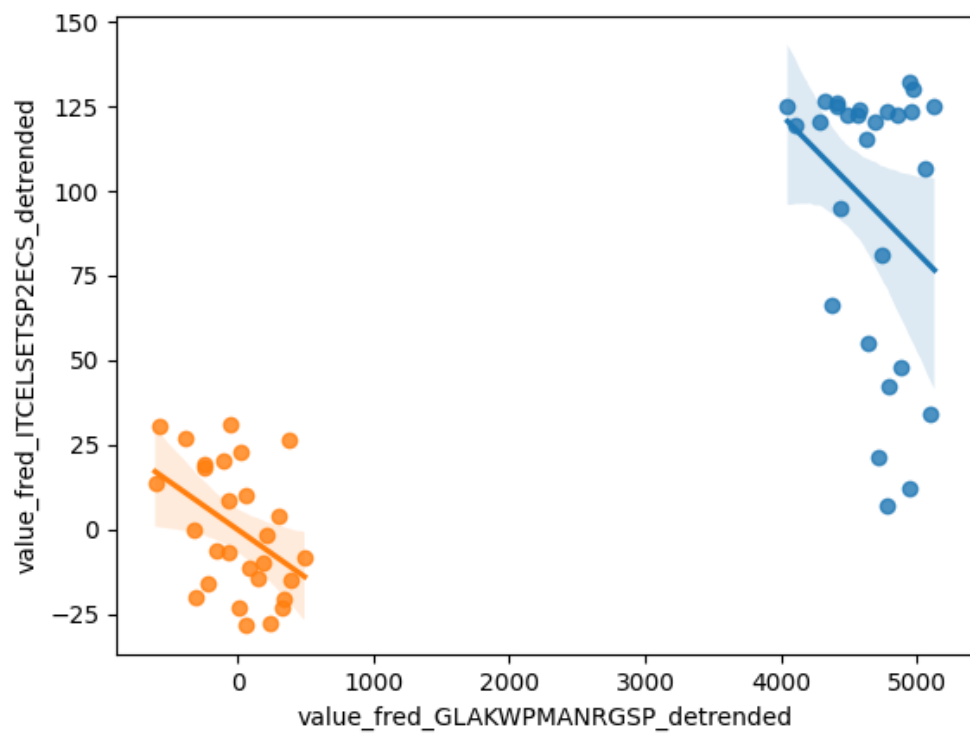


Figure 2: Detrended Regression Plot for 2026-09-28